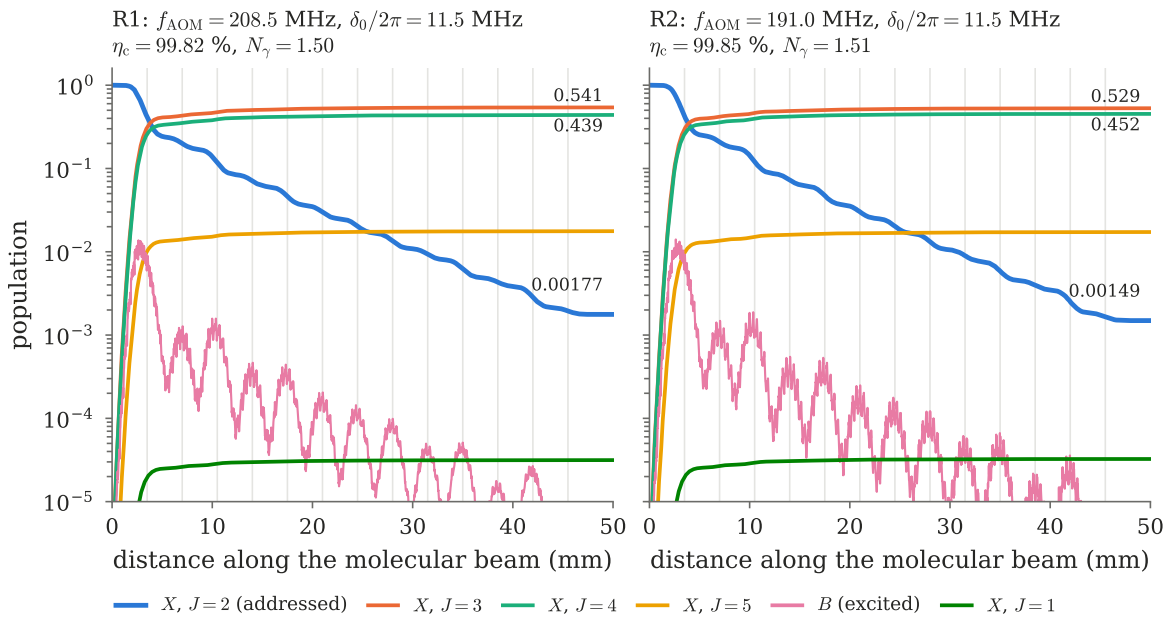

R1: $f_{\mathrm{AOM}} = 208.5$ MHz, $\delta_0/2\pi = 11.5$ MHz
$\eta_c = 99.82$ %, $N_\gamma = 1.50$
R2: $f_{\mathrm{AOM}} = 191.0$ MHz, $\delta_0/2\pi = 11.5$ MHz
$\eta_c = 99.85$ %, $N_\gamma = 1.51$
population
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0.541
0.439
0.00177
0.529
0.452
0.00149
0
10
20
30
40
50
distance along the molecular beam (mm)
distance along the molecular beam (mm)
X, $J = 2$ (addressed)
X, $J = 3$
X, $J = 4$
X, $J = 5$
B (excited)
X, $J = 1$